

Flow B - AWS Identity Containment with Native TheHive 5 Case Promotion

Security Hub custom action driven containment, Slack notification, DataTable state update, and promotion of the matching alert into a case

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Flow B AWS Identity Containment TheHive5 Native

Document Type

Workflow project document | Flow B

Prototype Focus

AWS IAM automation | n8n | TheHive 5 | Slack

Project Character

Portfolio prototype with implementation, testing, and validation evidence

Design Goal

Showcase an end-to-end SOC automation workflow with clear business and analyst value

Why this document matters

- This write-up is designed to work both as a project portfolio document and as an implementation-facing prototype note.
- It explains the problem, the workflow design, the operational value to a SOC, and the validation evidence that proves the workflow worked in the lab.
- The screenshots included throughout the document are the actual workflow, node, Slack, DataTable, AWS, and TheHive validation artifacts collected during the build process.

Key outcomes

- Uses a custom action event to trigger access-key containment in AWS IAM.
- Updates the workflow state in Slack and DataTable after containment is executed.
- Searches for the matching TheHive alert from Flow A and promotes it to a case using a containment-focused case template.

1. Project Summary

Flow B is the response and escalation workflow of the prototype. It begins after a finding has already been triaged and an analyst or custom action requests containment. The primary action implemented in the prototype is disabling the impacted IAM access key by setting it inactive through AWS IAM.

What makes Flow B especially valuable is that it does more than perform the cloud action. It also updates the operational record, notifies Slack, and carries the case-management context forward by promoting the matching TheHive alert into a fully tracked case with a case template and a case comment.

2. Problem Statement

A triaged identity finding is still not a contained incident. Without a response workflow, analysts must manually disable the affected key, update their chat tools, update tracking records, find the matching alert in TheHive, create or promote the case, and document what happened. That is slow, repetitive, and error-prone - especially when time matters.

Why the problem matters in a SOC

- Containment actions should be fast and consistent once the decision to act has been made.
- Analysts need the response status written back into the systems they already use, not only executed silently in AWS.
- The hand-off from alerting to case management is critical for investigations that move beyond a simple one-click response.

3. Objective

- Receive a supported custom action request for AWS IAM access-key containment.
- Disable the targeted key through AWS IAM when required fields are present.
- Record the outcome in Slack and the n8n DataTable.
- Find the matching TheHive alert and promote it to a case using a purpose-built case template.
- Create a case comment that documents the exact containment result.

4. Solution Overview

Flow B validates that the incoming message matches the expected containment action and that both the access key and principal context are present. When the request is supported, the workflow executes the AWS IAM update call that sets the key to Inactive. If required fields are missing or the action is unsupported, the workflow still records the status clearly for the analyst.

After the AWS action, the workflow builds TheHive context and searches for the matching alert that was originally created by Flow A. Once the alert is found, the workflow uses native TheHive 5 nodes to fetch the alert, promote it to a case with a containment case template, and add a case comment summarizing the exact result.

This makes Flow B a strong example of response orchestration because it combines a real cloud action with operational state management and investigation workflow continuity.

5. Tools and Technologies Used

Tool / Technology	Why it was used in this workflow
AWS IAM UpdateAccessKey	Executes the actual containment action by setting the targeted access key to Inactive.
AWS Security Hub custom action pattern	Provides a realistic response trigger pattern that can be tied to analyst approval or escalation logic.
n8n	Coordinates validation, branching, AWS action execution, state recording, and TheHive 5 integration.
TheHive 5 native nodes	Search, get, promote, and comment operations were implemented with native node actions after troubleshooting the HTTP approach.
TheHive case template	Standardizes the promoted case with predefined tasks and containment-focused handling guidance.
Slack and DataTable	Keep the containment result visible to analysts and preserve the response state in the project tracking table.

6. How the System Works

Stage	What happens
Custom action intake	Receives and normalizes the Security Hub custom action message.
Action validation	Checks that the requested action matches the supported containment path.
Key / principal checks	Confirms the workflow has enough context to execute a safe IAM action.
Containment execution	Calls IAM to set the targeted access key to Inactive.
Operational recording	Sends the result to Slack and updates DataTable.
TheHive continuation	Searches the matching alert, promotes it to a case, and adds a case comment.

Plain-English view

- In plain English, Flow B takes an already triaged identity alert and turns the response decision into action.
- It disables the key, tells the team what happened, and upgrades the alert into a case so investigation and closure can continue properly.

7. Workflow and Architecture Evidence

The images below show the final native-TheHive version of Flow B. The complete workflow view demonstrates the combined AWS response and TheHive case-management path, while the zoomed views highlight the containment segment and the later TheHive promotion branch.



Figure B1. Full Flow B workflow with containment and TheHive case promotion.

The workflow is easiest to understand when split into functional segments. The first zoom focuses on containment execution, while the second shows the TheHive search, promotion, and commenting path.



Figure B2. Flow B containment and status update segment.

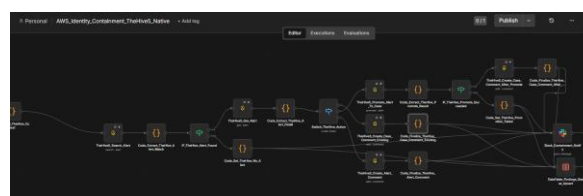


Figure B3. Flow B TheHive promotion branch.

The build process originally encountered issues with TheHive search and promotion through raw HTTP. The final design uses native TheHive 5 nodes, which improved reliability and made the implementation more maintainable.

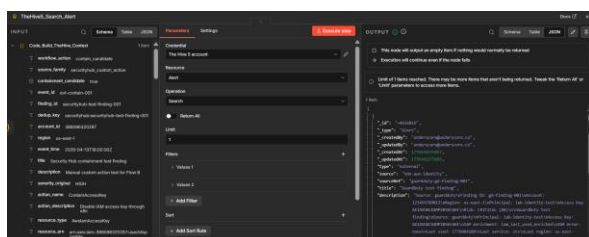


Figure B4. Native TheHive alert search configuration.

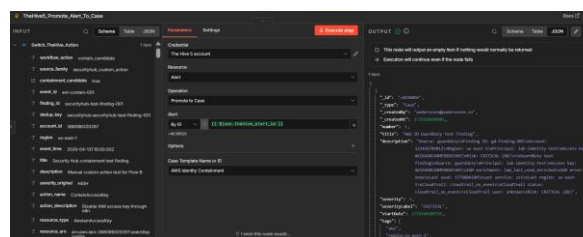


Figure B5. Native TheHive promote-to-case configuration with case template.

8. Key Features and USP

- Containment action matching and safety checks before AWS execution.
- Graceful branching for unsupported actions or missing context.
- Native TheHive 5 search, get, promote, and comment operations.
- Use of a dedicated containment case template for consistent case structure.
- Cross-platform visibility: AWS action result, Slack record, DataTable state, and TheHive case comment.

9. SOC Value and Expected Impact

Flow B reduces manual response friction after triage. In a real SOC, the value is not only the speed of disabling a key; it is the consistency of execution and the fact that the response evidence is automatically

propagated into collaboration and case-management systems. That improves auditability, response discipline, and follow-through after the initial action is taken.

Operational area	Expected effect
MTTR	Lower because the containment action is automated once the decision path is reached.
Response consistency	Improved because each containment event follows the same validated execution path.
Investigation continuity	Improved because the same alert is promoted into a templated case instead of starting from scratch.
Analyst workload	Reduced because manual key disable + manual case setup + manual documentation are minimized.
Case quality	Improved because the promoted case starts with predefined tasks and a containment note.

10. Validation and Testing Evidence

The first part of the validation confirmed that the AWS containment action itself worked. The Slack result below shows that the targeted key was set to Inactive and that the workflow recorded the containment summary clearly.

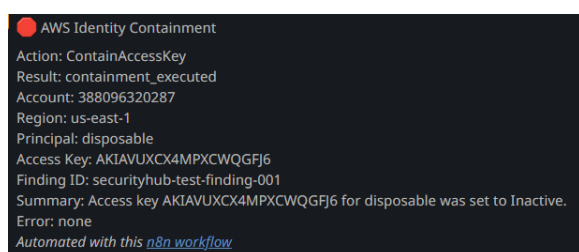


Figure B6. Slack validation showing the containment action executed.

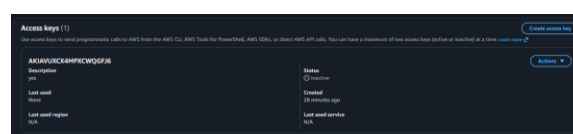


Figure B7. AWS IAM view showing the affected key after containment.

A major project milestone was the successful promotion of the Flow A alert into a TheHive case. The screenshots below show the resulting case list and the case detail with the generated comment.

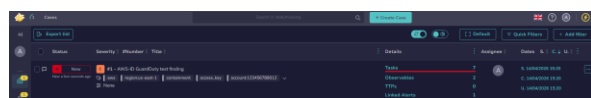


Figure B8. TheHive case list after alert promotion.

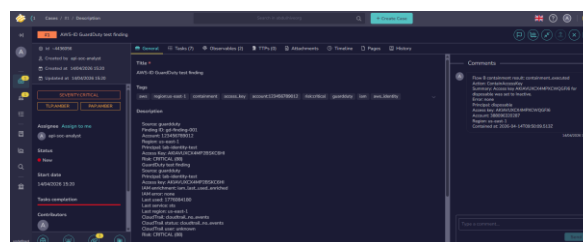


Figure B9. TheHive case detail with the containment comment.

The final analyst-facing validation in Slack included the alert reference, alert ID, case ID, alert URL, and case URL. This confirms that the response workflow was not only executed but also fully tied back into case management.

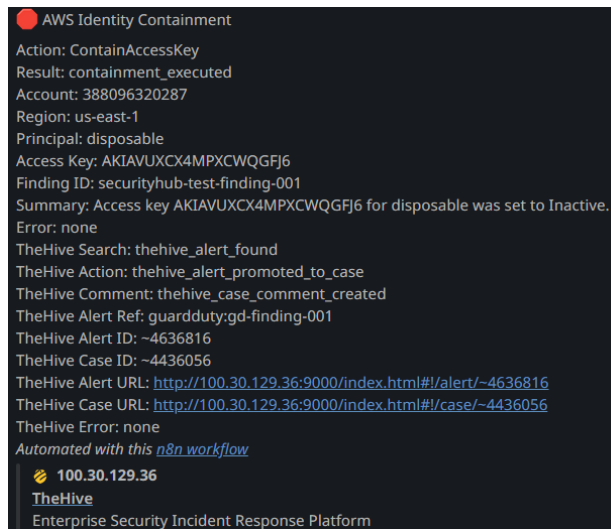


Figure B10. Final Slack output with TheHive alert and case references after successful promotion.

The dedicated case template used for promotion is also part of the project value because it standardizes the analyst work that follows containment.

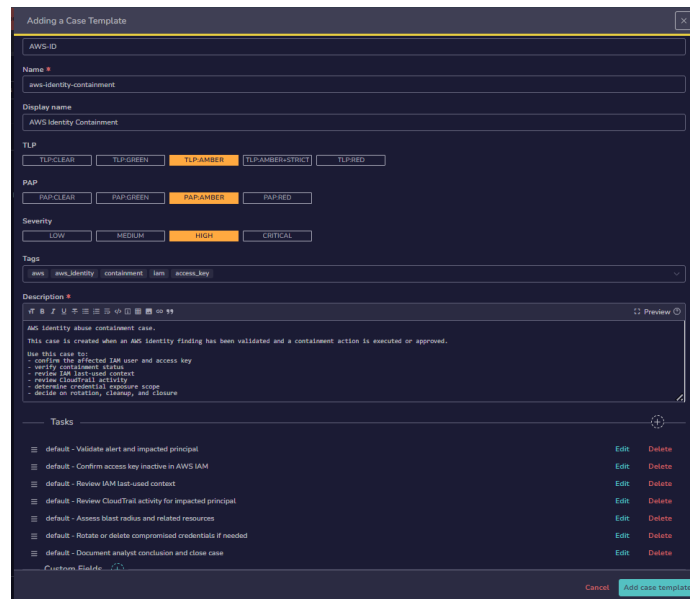


Figure B11. TheHive case template used for the promoted containment case.

11. Real-World Use Cases

- Compromised or suspicious IAM access keys that require fast deactivation.
- Analyst-approved response actions launched from Security Hub or another approval pattern.
- SOAR prototypes that must show both containment and investigation continuity.

12. Skills Demonstrated

- Cloud response automation using AWS IAM APIs.
- TheHive 5 case-management integration with native nodes.
- Workflow troubleshooting and redesign from HTTP-based approach to more reliable native integration.
- Operational documentation and state propagation across tools.

13. Prototype Limitations

- This prototype focuses on one containment action type: disabling an IAM access key.
- TheHive promotion depends on a matching alert already existing from Flow A.
- Production use would likely require stronger approval controls, RBAC, and guardrails around who can trigger containment.

14. Future Enhancements

- Add analyst approval or peer-review gates before privileged containment actions.
- Extend the response pattern to quarantine roles, sessions, or federated access paths.
- Push enrichment artifacts or evidence attachments directly into the promoted case.

15. Conclusion

Flow B proves the response layer of the prototype. It takes the project beyond alerting and demonstrates a realistic SOAR pattern: validated action execution in AWS, clear analyst visibility, durable state tracking, and promotion into structured case management. That combination makes it a strong portfolio example of cloud containment orchestration.

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